RASHMI KUMAR
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EDUCATION

Purdue University, Department of Chemistry
West Lafayette, Indiana 3.85/4.00
PhD in Chemistry June 2020

Indian Institute of Technology Madras
Chennai, India 8.59/10.00
Master of Science (M.Sc) in Chemistry June 2014

Presidency College (Calcutta University)
Kolkata, India 72.25/100
Bachelor of Science (B.Sc) in Chemistry May 2012

WORK EXPERIENCE

Postdoctoral Scholar May 2023 – current
Pacific Northwest National Lab
(Supervisor**: Dr. Ljiljana Paša-Tolić, Lab Fellow, Lead Scientist for Visual Proteomics,** EBSD Directorate)

- **Spatially resolved single cell analyses of proteomic and metabolomic changes associated with senescence in Idiopathic Pulmonary Fibrotic Lungs and Precision Cut Lung Slices (PCLS).**
  - Discovery of novel protein biomarkers associated with senescence in fibrotic lungs.
  - Combinatorial analysis of proteins and metabolites for understanding of cellular pathways associated with inflammatory senescence.
  - Multimodal spatial investigation of transcripts, glycans, extracellular matrix and proteins tissue microenvironment associated with developmental lung injury. ***(funded by NHLBI-LungMAP Trainee Award)***

- **Spatial multimodal analysis of Pancreatic Cancer Subtypes.**
  - Integrated proteomic and transcriptomic analyses of archival tissues, isolated from patients suffering from pancreatic adenosquamous carcinoma and pancreatic ductal adenocarcinoma.
  - Discovery of novel protein biomarkers differentiating pathological adenoma and squamous cells in pancreatic cancer.

- **Spatial cell-type resolved multimodal analysis of Renal Cell Carcinoma**
  - Spatial analysis of distinct proteoforms and post translational modifications in human kidney
  - Spatial mapping of diverse cell types associated with renal cell carcinoma
  - Identification of distinct proteins and glycosylation signatures, associated with various cell types in renal cancer.

Postdoctoral Scholar July 2020 – May 2023
Stanford University School of Medicine (*SoM Dean's Postdoctoral Fellowship*)
(Supervisor: **Dr. Mike Angelo,** Associate Professor in the Department of Pathology)

- **Investigation of role of chromosomal copy number changes associated with Breast Cancer *(in collaboration with Rob West, Professor at Stanford School of Medicine).***
  - Development of mass spectrometric imaging methods for spatial detection of chromosomal copy number changes in tissues using multiplexed ion beam imaging platform. (MIBI-TOF).

- Development of novel technique for mass-spectrometric multiplexed imaging of RNA and proteins simultaneously in FFPE tissues.
- Combinatorial analysis of chromosomal copy number and proteins identified distinct tissue microenvironments in high-risk breast cancer.

- **Development of techniques aimed towards improvement and enhancement of the multiplex capabilities of MIBI and CyTOF *(in collaboration with Sean Bendall, Associate Professor at Stanford School of Medicine).***
  - Developed a new polymer that can be conjugated to various bifunctional chelators suitable for various lanthanides and metals, rarely found in biological samples.
  - Improved multiplexing capability of MIBI-TOF assays by expanding the number of metals that can be conjugated to antibodies and detected simultaneously in human archival tissues.

Research Assistant August 2014 – June 2020
Purdue University
(Supervisor: **Prof. Hilkka I. Kenttämaa,** Frank Brown Distinguished Professor of Chemistry**)**

- **Generation and characterization of radical cations in solution and gas phase *(funded by NSF).***
  - Developed a method for distinguishing between exo- and endothermic reactions in the gas phase in linear quadrupole ion traps and used it for the experimental determination of the proton affinities of isomeric dehydro and didehydropyridines in the gas phase.
  - Generated *para*-benzyne biradicals in solution and established correlations between their gas-phase and solution reactivities.
  - Investigated nucleophilic addition reactions of para-benzynes in solution.

- **Development of mass spectrometric methods for the identification of polyfunctional drug metabolites via ion-molecule reactions.**
  - Developed mass spectrometric methods based on gas-phase ion-molecule reactions for the identification and characterization of drug metabolites.
  - Investigated and identified the protonation sites of polyfunctional analytes in gas-phase by using various dissociation techniques and gas-phase ion-molecule reactions.
  - Measured the lowest limit of detection for identification of drug metabolites using LC-MS coupled with gas-phase ion-molecule reactions.

Research Intern June 2018 – Dec 2018
Nestlé Development Centre, Marysville
(Supervisor: **Pamela Gilliland, Science and Technology Team Lead**)

- **Development of an automated sampling system.**
  Developed and optimized an automated sampling system that is time-efficient, robust, and broadly applicable to a variety of liquid samples and beverages.
- **Analysis of beverage aroma by using GC-MS.**
  Performed comprehensive analysis of beverage volatile compounds utilizing gas chromatography–mass spectrometry (GC-MS), employing both static and dynamic headspace sampling to achieve separation and quantification of aroma constituents
- **Application of novel microextraction techniques for the analysis of volatile aroma by using GC-MS/MS.**
  Applied advanced microextraction approaches—including solid-phase microextraction (SPME), stir bar sorptive extraction (SBSE), and thin-film SPME—coupled with GC-MS/MS, to enhance the analysis of volatile aroma compounds across diverse matrices.
- **GC-Olfactometric analysis of beverage aroma.**

Conducted GC-olfactometric analyses by integrating gas chromatographic separation with olfactometry and mass spectrometry, thus correlating the chemical identity of aroma compounds with their sensory attributes.

Research Assistant May 2013 – May 2014
Indian Institute of Technology, Madras, India
(Supervisor: **Prof. Sundarababu Baskaran,** Professor of Chemistry**)**

- **Development of stereoselective reactions at room temperature.**
  Development of new methodology for acetal formation at room temperature that can be applied to labile polyfunctional compounds. Selective generation of chiral centers for asymmetric molecules.
- **Organic reactions in low-melting mixtures.**
  Investigated acidic eutectic melts that are stable at room temperature and implemented acidic melting mixtures to facilitate various acid catalyzed reactions.
- **Synthesis of carbohydrates from a dioxanone scaffold using stereoselective Aldol condensation reactions.**
  Synthesized carbohydrates with multiple stereocenters from simpler molecules by generating chiral centers via single-step reactions, such as Aldol condensation.

TEACHING EXPERIENCE

Purdue University

- Teaching assistant for a graduate level **mass spectrometry** course; taught fundamental aspects of various mass spectrometric methods (**Chemical Aspects of Mass Spectrometry, CHM 622).**
- Teaching assistant for an undergraduate level **organic chemistry** course; demonstrated students the use of various laboratory techniques, such as IR Spectroscopy, Thin Layer Chromatography, and Gas Chromatography and taught students how to perform organic synthesis **(CHM 255).**
- Teaching assistant for an undergraduate level **organic chemistry** course; taught mechanisms of organic reactions **(CHM 257).**

CERTIFICATIONS

Mini-MBA degree: Krannert School of Business and Management, Purdue University, applied management studies, June 2018.

AWARDS AND SCHOLARSHIPS

- NHLBI-LungMAP Trainee Pilot Grant Award, 2025 for project titled: Multimodal Investigation of the Tissue Microenvironment in Developmental Lung Injury.
- US HUPO Travel grant award 2024 for presentation of project titled, “Spatial Mapping of Proteomic Changes Associated with Senescence in Idiopathic Pulmonary Fibrosis”.
- Females in Mass Spectrometry Empowerment Award, Q3-2022.
- Young Investigator Education Grant 2022 by the Mass Spectrometry & Advances in the Clinical Lab conference.
- Stanford School of Medicine Dean’s Postdoctoral Fellowship 2021.
- Robert R. Squires Scholarship Award 2017 by the Department of Chemistry, Purdue University.
- H.C. Brown Organic Division Travel Grant Award 2016 by the Department of Chemistry, Purdue University.
- INSPIRE Scholarship offered by the Central Government of India (2009-2014).
- Recipient of a scholarship offered by West-Bengal Board of Higher Secondary Education for ranking 69 in Class-XII finals (Higher Secondary Examination) in the state of West Bengal: offered to top 10 % high-school students in the merit list of the final year all-state exam.

PUBLICATIONS

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- *Hierarchical blade architecture in a multi-tiered µDicer enables mechanical dissection of tissues down to 10 µm resolution***,** Annatoma Arif, **Rashmi Kumar,** Yumi Kwon, Seth C. Cordts, Saisneha Koppaka, Ying Zhu, Ljiljana Paša-Tolić, and Sindy K. Y. Tang, *Small (2026): e14816*
- *Unsupervised semantic segmentation models for tissue region of interest identification in histology images*, David J. Degnan, Bailey G. Knight, Logan A. Lewis, Ian A. VanGordon, Teminijesu T.Adewunmi, Lauren M. Jones, Shruthilayaa Sivakumar, **Rashmi Kumar**, Dušan Veličković, Kevin J. Zemaitis, Ljiljana Paša-Tolic, Lisa M. Bramer**,** (**accepted** in Journal of the American Society for Mass Spectrometry).
- *New Atomic Mass Tags for Enhanced Multiplexing Capability of Multiplexed Ion Beam Imaging Time-of-Flight (MIBI-TOF) Analysis***, Rashmi Kumar**, Felix J. Hartmann, Patricia Favaro, Daniel Ho, Trevor Bruce, Mako Goldston, Angie Spence, Erin F. McCaffrey, Sean C. Bendall, and Michael Angelo, Analytical Chemistry*,* **2025** *97* (15), 8220-8230.
- *Spatial Top-down Proteomics for the Functional Characterization of Human Kidney***, Rashmi Kumar**, Kevin J. Zemaitis, James M Fulcher, David J. Degnan, Logan A. Lewis, Yen-Chen Liao, Ljiljana Paša-Tolic, Clinical Proteomics, **22**, 9 (2025).
- *Multi-omic landscape of human gliomas from diagnosis to treatment and recurrence***,** Hadeesha Piyadasa, Benjamin Oberlton, Mikaela Ribi, Jolene S Ranek, Inna Averbukh, Ke Leow, Meelad Amouzgar, Candace C Liu, Noah F Greenwald, Erin F McCaffrey, **Rashmi Kumar**, Selena Ferrian, Albert G Tsai, Ferda Filiz, Christine Camacho Fullaway, Marc Bosse, Sricharan Reddy Varra, Alex Kong, Cameron Sowers, Melanie Hayden Gephart, Pablo Nuñez-Perez, EnJun Yang, Mike Travers, Michael J Schachter, Samantha Liang, Maria R Santi, Samantha Bucktrout, Pier Federico Gherardini, John Connolly, Kristina Cole, Michael E Barish, Christine E Brown, Derek A Oldridge, Richard R Drake, Joanna J Phillips, Hideho Okada, Robert Prins, Sean C Bendall, Michael Angelo*,(accepted),* Cancel Cell. doi: 10.1101/2025.03.12.642624. PMID: 40161803; PMCID: PMC11952471.
- *Advances in Mass Spectrometry-enabled Multiomics Analyses at Single-Cell Resolution,* **Rashmi Kumar**, Kevin J. Zemaitis, James M Fulcher, Ljiljana Paša-Tolic, Current Opinions in Biotechnology, 2024, 87, 103096.
- *Synthesis, Characterization, and Applications of a Superior Dendrimer-Based Polymer for Multiplexed Ion Beam Imaging Time-of-Flight Analysis***, Rashmi Kumar**, Candace C Liu, Sean C. Bendall, Michael Angelo, Biomacromolecules 2023, **24**, 7, 3105-3114.
- *Differentiation of the Aziridine Functionality from Related Functional Groups in Protonated Analytes by Using Selective Ion-Molecule Reactions Followed by Collision-Activated Dissociation in a Linear Quadrupole Ion Trap Mass Spectrometer***,** Ruth O. Anyaeche, Kawthar Z. Alzarieni, **Rashmi Kumar**, Jacob R. Milton, Jaskiran Kaur, Huaming Sheng, and Hilkka I. Kenttämaa, The Journal of Organic Chemistry 2023, **88**, 8865-8873.
- *The immunoregulatory landscape of human tubercuosis granulomas***,** Erin F. McCaffrey, Michele Donato, Leeat Keren, Zhenghao Chen, Alea Delmastro, Megan B. Fitzpatrick, Sanjana Gupta, Noah F. Greenwald, Alex Baranski, William Graf, **Rashmi Kumar**, Marc Bosse, Christine Camacho Fullaway, Pratista K. Ramdial, Erna Forgó, Vladimir Jojic, David Van Valen, Smriti Mehra, Shabaana A. Khader, Sean C. Bendall, Matt van de Rijn, Daniel Kalman, Deepak Kaushal, Robert L. Hunter, Niaz Banaei, Adrie J. C. Steyn, Purvesh Khatri and Michael Angelo, Nature Immunology 2022, **23,** 318–329.
- *Multiplexed imaging reveals an IFN-g-driven inflammatory state in nivolumab-associated gastritis*, Selena Ferrian, Candace C. Liu, Erin F. McCaffrey, **Rashmi Kumar**, Theodore S. Nowicki, David W. Dawson, Alex Baranski, John A. Glaspy, Antoni Ribas, Sean C. Bendall, and Michael Angelo, Cell Reports Medicine 2021, (10) 2, 100419.
- *Measurements of the Proton Affinities of a Series of Mono- and Biradicals of Pyridine,* **Rashmi Kumar**, Guannan Li, Vanessa A. Gallardo, Anyin Li, Jacob Milton, John J. Nash, and Hilkka I. Kenttämaa, Journal of the American Chemical Society 2020, (142), 8679-8687.
- *Effects of Residual Water in a Linear Quadrupole Ion Trap on the Protonation Sites of 4-Aminobenzoic Acid,* **Rashmi Kumar**, Ravikiran Yerabolu, and Hilkka I. Kenttämaa, Journal of the American Society for Mass Spectrometry 2020, (31), 124-131.
- *The Effects of Analyte Concentration on the Protonation Sites of 4-Aminobenzoic Acid upon Atmospheric Pressure Chemical Ionization as Revealed by Gas-Phase Ion-Molecule Reactions,* **Rashmi Kumar** and Hilkka I. Kenttämaa, Journal of the American Society for Mass Spectrometry 2020, (31), 2210-2217.
- *A Direct Method for the Efficient Synthesis of Benzylidene Acetal at Room Temperature,* Rajashekar Narra Reddy, **Rashmi Kumar**, and Sundarababu Baskaran, European Journal of Organic Chemistry 2019 (7), 1548-1552.
- *Factors Affecting the Limit of Detection for HPLC/Tandem Mass Spectrometry Experiments Based on Gas-Phase Ion-Molecule Reactions***,** Leah F. Easterling, Ravikiran Yerabolu, **Rashmi Kumar**, Kawthar Alzarieni, and Hilkka I. Kenttämaa, Analytical Chemistry 2020**,** 92 (11), 7471-7477.

- *Gas Phase Fragmentation of Adducts between Dioxygen and closo-Borate Radical Anions*, Jonas Warneke,∗, Markus Rohdenburg, Judy K.Y. Liu, Erynn Johnson, Xin Ma, **Rashmi Kumar**, Pei Su, Edoardo Aprà, Xue-Bin Wang, Carsten Jenne, Maik Finze, Hilkka I. Kenttämaa, and Julia Laskin**,** International Journal of Mass Spectrometry, 2019 (436), 71-78.
- *Integration of a Multichannel Pulsed-Valve Inlet System to a Linear Quadrupole Ion Trap Mass Spectrometer for the Rapid Consecutive Introduction of Nine Reagents for Diagnostic Ion/Molecule Reactions,* John Y. Kong, Ryan T. Hilger, Chunfen Jin, Ravikiran Yerabolu, James R. Zimmerman, Randall W. Replogle, Tiffany M. Jarrell, Leah Easterling, **Rashmi Kumar**, and Hilkka I. Kenttämaa, Analytical Chemistry, 2019 (91), 15652-15660.

PREPRINTS

- *Heat-Induced Generation of para-Benzynes in Solution and Comparison of Their Reactivities in Gas Phase and Solution*, **Rashmi Kumar** and Hilkka I. Kenttämaa, (*in preparation*).
- *Evaluation and application of chemical decrosslinking in the context of histopathological spatial proteomics,* Andikan J. Nwosu, Liang Chen, **Rashmi Kumar**, Yumi Kwon, Shaun M. Goodyear, Adel Kardosh, James M. Fulcher and Ljiljana Paša-Tolic *(https://doi.org/10.64898/2026.02.06.704439)*
- Single-Cell Proteomics of Human Peripheral Blood Mononuclear Cells Exceeding 600 Cells per Day, James M Fulcher, Yumi Kwon, Pranav Dawar, **Rashmi Kumar**, Sarah M Williams, Patricia Miller, Andrey Liyu, Liang Chen, Daniel J Orton, Heather M Olson, Fengchao Yu, Alexey I Nesvizhskii, Julie Fortier, Ravi Vij, Reyka Jayasinghe, Li Ding, Ying Zhu, Ljiljana Pasa-Tolic *(https://doi.org/10.64898/2026.02.01.703169)*

SCIENTIFIC CONTRIBUTIONS AND OUTREACH

- Reviewer for journal, Analytical Chemistry (2022-current).
- Reviewer for journal, International Journal of Mass Spectrometry (2022-current).
- Co-chair of PNNL-SPARK employee resource group for promoting diversity and equity for women and gender-diverse staff at PNNL and across national labs (2023-2024).
- Member of Stanford Pathology DE&I Committee.
- Volunteer at Science is Elementary (SiE) aimed towards providing hands-on experience with scientific experiments to under-privileged children.

MENTORING AND LEADERSHIP SKILLS

- Trained younger graduate students in maintenance and trouble-shooting Thermo Scientific Linear Quadrupole Ion Traps and Orbitraps
- Mentored graduate students on gas-phase ion-molecule reactions
- Taught an undergraduate student the basics of organic synthesis and supervised the synthesis of lignin polymers
- In charge of the maintenance and trouble-shooting of LQIT and high resolution Orbitraps.
- In charge of safety for a research group
- In charge of safe storage and safe handling of chemicals for a research group and of the group chemical inventory

SEMINARS/CONFERENCES

- Oral presentation: '**Spatial Mapping of Proteomic Changes Associated with Senescence in Idiopathic Pulmonary Fibrosis'**, **ASMS** *2024*, Anaheim, California
- Poster presentation: **'A Novel Ion-beam Imaging Technique for Detection of Chromosomal Copy Number Alterations Associated with Progression of Breast Cancer', *ASMS 2022***, Minneapolis, Minnesota.
- Poster presentation: **'A Novel Ion-beam Imaging Technique for Detection of Chromosomal Copy Number Alterations Associated with Progression of Breast Cancer', *MSACL 2022***, Monterey, California.
- Poster presentation: **'A Novel Ion-beam Imaging Technique for Detection of Chromosomal Copy Number Alterations Associated with Progression of Breast Cancer', *US HUPO 2022***, Charleston, South Carolina.
- Poster presentation: '**Studying the Effects of Analyte Concentration on the Protonation Sites of 4-Aminobenzoic Acid Using Gas-phase Ion-molecule Reactions', *ASMS*** 2018, San Diego, California.

- Poster presentation: '**A Comparison of Reactivity of *para*-Benzynes in Gas Phase and Solution', *ASMS*** 2017, Indianapolis, Indiana.
- Poster Presentation: **'Study of Protonation Sites in Multifunctional Analytes Upon Positive Mode ESI and APCI', *Gordon Research Conference: Gaseous Ions, 2017***, Ventura, California
- Poster presentation: '**Protonation Sites in Multifunctional Compounds Upon Positive Mode ESI and APCI in a Linear Quadrupole Ion Trap and Orbitrap', *ASMS*** 2016, San Antonio, Texas.
- Attended the ***'ACS on Campus'*** conference at IIT Madras in November 2013.